Appendix E. Perplexity Under Steering

We found that steering with magnitudes of |1.5| for last prompt token derived directions and even at |1.0| for response token derived directions increased perplexity dramatically. Following are some tables displaying the perplexity of the model on WikiText-2 under steering.

Mech.	Direction, vector	−0.5	−1	−1.5
fabrication	instructed, prompt	5.23	7.28	13.85
	instructed, response	7.99	48.17	276
	spontaneous, prompt	5.46	8.11	14.64
	spontaneous, response	7.62	44.13	401
omission	instructed, prompt	5.25	7.32	13.44
	instructed, response	6.95	13.95	63.55
	spontaneous, prompt	5.64	8.54	17.92
	spontaneous, response	7.95	64.51	1313
random (6 draws)		4.86–4.90	5.12–5.21	5.67–5.85
unsteered baseline		4.81	4.81	4.81

Table 13: Perplexity scores on wikitext under suppression steering at layers 19-32 over all tokens. Steering with random vectors leaves perplexity near baseline. Response token derived direction steering dramatically increases perplexity scores, while also steering less effectively than last prompt token derived directions.

Mech.	Direction, vector	+0.5	+1	+1.5
fabrication	instructed, prompt	5.78	9.12	15.84
	instructed, response	6.23	14.52	50.05
	spontaneous, prompt	5.34	7.65	15.22
	spontaneous, response	5.52	9.35	35.50
omission	instructed, prompt	5.56	8.19	14.05
	instructed, response	5.61	11.29	79.64
	spontaneous, prompt	5.35	7.71	15.89
	spontaneous, response	6.16	56.78	11146
random (6 draws)		4.89–4.94	5.19–5.32	5.81–6.09
unsteered baseline		4.81	4.81	4.81

Table 14: Perplexity scores on WikiText-2 under induction steering at layers 19-32 over all tokens. Steering with random vectors leaves perplexity near baseline. Response token derived direction steering dramatically increases perplexity scores, while also steering less effectively than last prompt token derived directions.

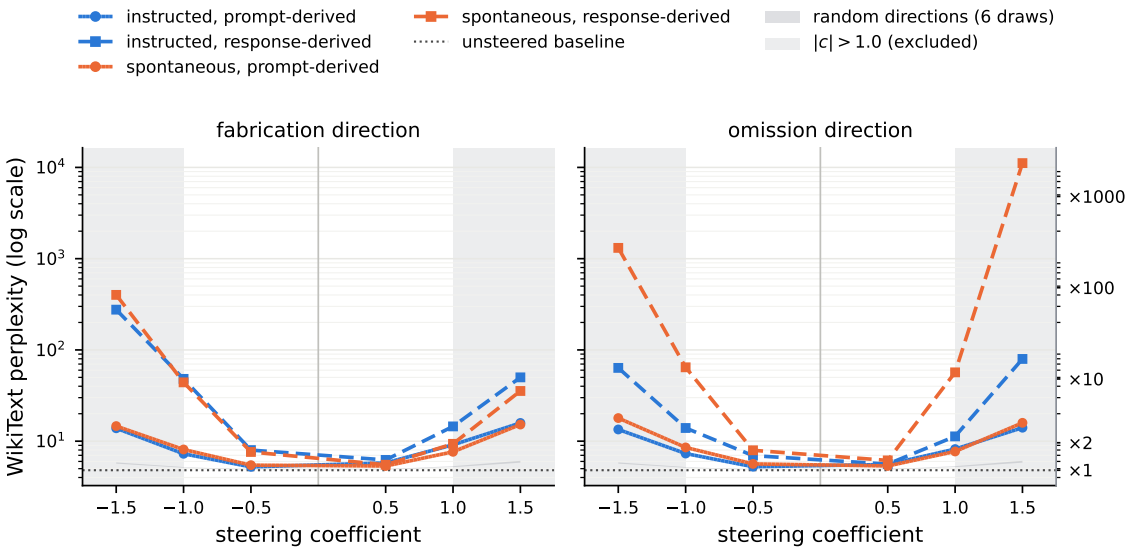


Figure 8: Perplexity scores on WikiText-2 under steering at layers 19-32 for all tokens. The y axis is logarithmic due to the dramatic scores some direction steering produced. The right hand axis gives multiples of the unsteered baseline.

Appendix F. Reproducibility

The repository for this paper can be found at https://github.com/JosiahL98/Spont-Instructed-Deception. The scaffolding for our repository builds on the codebase of Yang et al. (2024).

Experiments were run through NDIF, the OpenAI API, and OpenRouter API. The model studied was Llama-3.1-70B-Instruct, 80 layers, d model 8192. Residual stream activations were extracted through NNsight and NDIF. Greedy decoding was used throughout in generation by the model.

Gpt-5-mini was used via OpenRouter to generate the spontaneous and instructed deception prompts. Gpt-4.1-mini-2025-04-14 was used via the OpenAI API to generate the ratings of deception and coherence for the Model's responses to the prompts.

Tables and figures in this paper were generated by scripts available in the repository. The mapping from data to tables and figures is listed in the paper runs manifest file in the repo.

Random seeds 43 and 44 were used in addition to 42 at layer 35 in probing experiments. Random directions were computed with seeds 0, 1, and 2. All other code using random seeds used seed 42.